

1. Applying Perimeter Calculation in Contour Representation

A computer vision system is used to inspect rectangular metal plates in a factory. The system extracts the outer contour of a plate to measure its boundary. The detected rectangular plate has a width of 80 pixels and a height of 40 pixels. The boundary length is used as a contour-based feature. Apply the perimeter formula and calculate the contour length.

- A. 120 pixels
- B. 160 pixels
- C. 200 pixels
- D. 240 pixels

Correct Answer: D. 240 pixels

Explanation:

$$\text{Perimeter} = 2(\text{Width} + \text{Height})$$

$$\text{Perimeter} = 2(80 + 40)$$

$$\text{Perimeter} = 2 \times 120 = 240$$

So, the contour length is 240 pixels.

2. Applying Area Calculation from a Closed Contour

A medical image analysis system detects the contour of a lesion in a skin image. The detected lesion is approximately rectangular in shape. Its width is 30 pixels and height is 20 pixels. The system calculates the area enclosed by the contour to estimate lesion size. Apply the area formula and find the enclosed region area.

- A. 300 pixels²
- B. 500 pixels²
- C. 600 pixels²
- D. 900 pixels²

Correct Answer: C. 600 pixels²

Explanation:

$$\text{Area} = \text{Width} \times \text{Height}$$

$$\text{Area} = 30 \times 20 = 600$$

So, the area enclosed by the contour is 600 pixels².

3. Applying Aspect Ratio as a Shape Feature

An object recognition system extracts the contour of a road sign. The bounding box of the detected contour has width 120 pixels and height 60 pixels. The system uses aspect ratio as a contour-based shape descriptor to help classify the sign. Apply the aspect ratio formula and find the value.

- A. 0.5
- B. 1.0
- C. 1.5
- D. 2.0

Correct Answer: D. 2.0

Explanation:

$$\text{Aspect Ratio} = \frac{\text{Width}}{\text{Height}}$$

$$\text{Aspect Ratio} = \frac{120}{60} = 2.0$$

So, the aspect ratio is 2.0.

4. Applying Circularity Measure

A quality inspection system checks whether bottle caps are circular. The detected contour has an area of 314 pixels² and perimeter of 62.8 pixels. Apply the formula and calculate the approximate circularity.

- A. 0.25
- B. 0.50
- C. 0.75
- D. 1.00

Correct Answer: D. 1.00

Explanation:

Circularity is calculated using:

$$\text{Circularity} = \frac{4\pi A}{P^2}$$

where A is area and P is perimeter. A value close to 1 indicates a circle.

$$\text{Circularity} = \frac{4\pi \times 314}{(62.8)^2}$$

Using $\pi \approx 3.14$:

$$\text{Circularity} = \frac{4 \times 3.14 \times 314}{3943.84}$$

$$\text{Circularity} = \frac{3943.84}{3943.84} = 1.00$$

So, the contour is approximately circular.

6. Applying Compactness Feature

A leaf classification system extracts the contour of a leaf image. The leaf region has area 100 pixels² and perimeter 40 pixels. Apply the compactness formula and find the value.

- A. 8
- B. 12
- C. 16
- D. 20

Correct Answer: C. 16

Explanation:

The system calculates compactness using:

$$Compactness = \frac{P^2}{A}$$

where P is perimeter and A is area. Compactness is used to describe how spread out the shape boundary is.

$$\begin{aligned} Compactness &= \frac{P^2}{A} \\ Compactness &= \frac{40^2}{100} \\ Compactness &= \frac{1600}{100} = 16 \end{aligned}$$

So, the compactness value is 16.

1. Applying Area Calculation in Region-Based Representation

A medical image analysis system segments a tumor region from an MRI scan. Instead of using only the boundary, the system analyzes all pixels inside the detected tumor region. The segmented region contains 750 pixels, and each pixel represents 0.4 mm² in the actual image. The doctor wants to estimate the tumor area. Apply region-based representation and calculate the tumor area.

- A. 250 mm²
- B. 300 mm²
- C. 350 mm²
- D. 400 mm²

Correct Answer: B. 300 mm²

Explanation:

$$\begin{aligned} \text{Area} &= \text{Number of Pixels} \times \text{Area per Pixel} \\ \text{Area} &= 750 \times 0.4 = 300 \text{ mm}^2 \end{aligned}$$

So, the tumor area is 300 mm².

2. Applying Region Pixel Count

A satellite image processing system segments a water body from an aerial image. The detected water region contains 1200 blue pixels and 300 dark boundary pixels. Since region-based representation considers the complete interior region, both types of pixels are included in the region analysis. The system wants to calculate the total region size. Apply region-based representation and find the total number of pixels in the water region.

- A. 900 pixels
- B. 1200 pixels
- C. 1500 pixels
- D. 1800 pixels

Correct Answer: C. 1500 pixels

Explanation:

$$\begin{aligned} \text{Total Region Pixels} &= \text{Interior Pixels} + \text{Boundary Pixels} \\ \text{Total Region Pixels} &= 1200 + 300 = 1500 \end{aligned}$$

So, the total water region size is 1500 pixels.

3. Applying Mean Intensity of a Region

A computer vision system analyzes a segmented road region in a traffic image. The gray-level intensities of five pixels inside the region are 40, 50, 60, 70, and 80. The system calculates the mean intensity as a region-based feature. This helps compare road regions under different lighting conditions. Apply the mean intensity formula and find the region feature value.

- A. 50
- B. 55
- C. 60
- D. 65

Correct Answer: C. 60

Explanation:

$$\begin{aligned} \text{Mean} &= \frac{40 + 50 + 60 + 70 + 80}{5} \\ \text{Mean} &= \frac{300}{5} = 60 \end{aligned}$$

So, the mean intensity of the region is 60.

4. Applying Region Density Calculation

A smart agriculture system identifies a diseased leaf region from a plant image. The bounding box around the diseased region contains 1000 pixels. The actual diseased pixels inside the bounding box are 650 pixels. Apply the formula and find the region density.

- A. 0.45
- B. 0.55
- C. 0.65
- D. 0.75

Correct Answer: C. 0.65

Explanation:

The system calculates region density as:

$$Density = \frac{Region\ Pixels}{Bounding\ Box\ Pixels}$$

$$Density = \frac{650}{1000} = 0.65$$

So, the region density is 0.65.

5. Applying Centroid Calculation of a Region

A robot vision system detects a small object region using image segmentation. Three pixels belonging to the region have coordinates (2, 4), (4, 6), and (6, 8). The robot calculates the centroid of the region to identify the object's center position. Apply the centroid formula and find the centroid coordinate.

- A. (3, 5)
- B. (4, 6)
- C. (5, 7)
- D. (6, 8)

Correct Answer: B. (4, 6)

Explanation:

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4$$
$$\bar{y} = \frac{4 + 6 + 8}{3} = 6$$

So, the centroid is: (4, 6)

6. Applying Region-Based Classification Using Area

A fruit sorting machine segments fruit regions from camera images. A fruit region is classified as “small” if its area is below 500 pixels, “medium” if its area is between 500 and 1000 pixels, and “large” if its area is above 1000 pixels. One segmented fruit region contains 850 pixels. The system must classify the fruit size. Apply the given classification rule and identify the fruit size.

- A. Small
- B. Medium
- C. Large
- D. Not a fruit

Correct Answer: B. Medium

Explanation:

Given area:

850 pixels

Since:

$$500 \leq 850 \leq 1000$$

The fruit is classified as medium.

1. Applying Deformable Curve Energy Calculation

A medical image analysis system uses a deformable curve to detect the boundary of a tumor in an MRI image. The curve changes its shape gradually until it fits the tumor boundary. At one iteration, the internal energy of the curve is 12, and the external image energy is 18. The total energy is calculated as the sum of internal and external energies. Apply the energy calculation and find the total energy of the deformable curve.

- A. 20
- B. 25
- C. 30
- D. 35

Correct Answer: C. 30

Explanation:

$$\begin{aligned} \text{Total Energy} &= \text{Internal Energy} + \text{External Energy} \\ \text{Total Energy} &= 12 + 18 = 30 \end{aligned}$$

So, the total energy is 30.

3. Applying Boundary Length Change

A computer vision system uses a deformable curve to fit the outline of a leaf. Initially, the curve length is 150 pixels. After deformation, the curve fits the leaf boundary and its length becomes 180 pixels. The system calculates the change in curve length. Apply the length change calculation and find the increase in boundary length.

- A. 20 pixels
- B. 25 pixels
- C. 30 pixels
- D. 35 pixels

Correct Answer: C. 30 pixels

Explanation:

$$\begin{aligned} \text{Length Change} &= \text{Final Length} - \text{Initial Length} \\ \text{Length Change} &= 180 - 150 = 30 \end{aligned}$$

So, the boundary length increased by 30 pixels.

4. Applying Iterative Deformation Update

A deformable surface model is used to reconstruct the surface of an organ from 3D medical image data. At one vertex, the current position is $x = 20$. The image force moves the vertex by +3 units, and the smoothing force moves it by -1 unit. The new position is calculated by adding both force effects to the current position. Apply the update rule and find the new vertex position.

- A. 18
- B. 20
- C. 22
- D. 24

Correct Answer: C. 22

Explanation:

$$\begin{aligned} \text{New Position} &= \text{Current Position} + \text{Image Force} + \text{Smoothing Force} \\ \text{New Position} &= 20 + 3 - 1 = 22 \end{aligned}$$

So, the new vertex position is 22.

5. Applying Surface Area Approximation

A deformable surface is used to model a damaged region on a 3D object. The surface is divided into small triangular patches. If each triangular patch has an area of 5 mm² and the surface contains 40 patches, the system estimates the total surface area. Apply the surface area calculation and find the total area.

- A. 100 mm²
- B. 150 mm²
- C. 200 mm²
- D. 250 mm²

Correct Answer: C. 200 mm²

Explanation:

$$\begin{aligned} \text{Total Surface Area} &= \text{Number of Patches} \times \text{Area per Patch} \\ \text{Total Surface Area} &= 40 \times 5 = 200 \text{ mm}^2 \end{aligned}$$

So, the total surface area is 200 mm².

1. Applying Total Energy Calculation in Active Contours

A medical image analysis system uses an active contour model to detect the boundary of a tumor in an MRI scan. The snake moves toward the tumor boundary by minimizing its energy. At one iteration, the internal energy is 15, the image energy is 25, and the constraint energy is 10. The total snake energy is calculated by adding all these energy terms. Apply the active contour energy model and calculate the total energy.

- A. 40
- B. 45
- C. 50
- D. 55

Correct Answer: C. 50

Explanation:

$$\begin{aligned} E_{total} &= E_{internal} + E_{image} + E_{constraint} \\ E_{total} &= 15 + 25 + 10 = 50 \end{aligned}$$

So, the total snake energy is 50.

3. Applying Internal Energy Weight

A snake model is used to detect the boundary of a road sign. The smoothness term of the snake is 30, and its weight is 0.4. The continuity term is 20, and its weight is 0.6. Apply the formula and find the internal energy.

- A. 20
- B. 22
- C. 24
- D. 26

Correct Answer: C. 24

Explanation:

The weighted internal energy is calculated as:

$$E_{internal} = \alpha E_{smoothness} + \beta E_{continuity}$$

$$E_{internal} = 0.4(30) + 0.6(20)$$

$$E_{internal} = 12 + 12 = 24$$

So, the internal energy is 24.

5. Applying Iterative Position Update

An active contour point is moving toward an object boundary. Its current position is $x = 50$. The image force moves it by +6 units, while the smoothness force pulls it back by -2 units. The new position is calculated by adding the force effects to the current position. Apply the update rule and find the new position.

- A. 52
- B. 54
- C. 56
- D. 58

Correct Answer: B. 54

Explanation:

$$\text{New Position} = \text{Current Position} + \text{Image Force} + \text{Smoothness Force}$$

$$\text{New Position} = 50 + 6 - 2 = 54$$

So, the new position is 54.

6. Applying Energy Reduction Check

A snake model is applied to segment a blood vessel boundary. At iteration 1, the total energy is 120. At iteration 2, the total energy becomes 90. The system checks how much the energy has reduced after one iteration. Energy reduction indicates that the snake is moving toward a better boundary fit. Apply energy reduction calculation and find the reduction value.

- A. 20
- B. 25
- C. 30
- D. 35

Correct Answer: C. 30

Explanation:

$$\text{Energy Reduction} = 120 - 90 = 30$$

So, the snake energy has reduced by 30.

1. Applying Level Set Sign Convention

A medical image segmentation system uses level set representation to detect a tumor boundary. The level set function $\phi(x, y)$ represents regions using sign values. In this system, $\phi < 0$ represents the inside of the object, $\phi = 0$ represents the boundary, and $\phi > 0$ represents the outside region. At one pixel, the level set value is $\phi = -5$. Apply the level set sign convention and identify the pixel location.

- A. Inside the object
- B. On the boundary
- C. Outside the object
- D. Background noise only

Correct Answer: A. Inside the object

Explanation:

Given:

$$\phi = -5$$

Since:

$$\phi < 0$$

the pixel lies inside the object region.

2. Applying Boundary Identification Using Zero Level Set

A computer vision system segments a circular object from an image. It uses a level set function where the object boundary is represented by the zero level set. A set of pixels has the following level set values: $-3, -1, 0, 2, 5$. The system must identify which value corresponds to the actual object boundary. Apply the level set boundary rule and choose the correct value.

- A. -3
- B. -1
- C. 0
- D. 5

Correct Answer: C. 0

Explanation:

In level set representation, the boundary is represented as:

$$\phi(x, y) = 0$$

Therefore, the value 0 indicates the object boundary.

3. Applying Distance from Boundary

A level set method is used to segment an organ in an ultrasound image. The level set function is initialized as a signed distance function. At a pixel, the value of the level set function is $\phi = 7$. The sign indicates the region, while the magnitude indicates distance from the boundary. Apply the signed distance concept and find the distance from the boundary.

- A. 0 pixels
- B. 3 pixels
- C. 5 pixels
- D. 7 pixels

Correct Answer: D. 7 pixels

Explanation:

For a signed distance function:

$$\begin{aligned} \text{Distance} &= |\phi| \\ \text{Distance} &= |7| = 7 \end{aligned}$$

So, the pixel is 7 pixels away from the boundary.

4. Applying Region Classification from Level Set Values

A level set segmentation algorithm is applied to a road image to separate a vehicle from the background. The system uses the convention: $\phi < 0$ means vehicle region, $\phi = 0$ means boundary, and $\phi > 0$ means background. Four pixels have values $-4, 0, 3, -2$. Apply the rule and find how many pixels belong to the vehicle region.

- A. 1
- B. 2
- C. 3
- D. 4

Correct Answer: B. 2

Explanation:

Vehicle region is represented by:

$$\phi < 0$$

The negative values are:

$$-4, -2$$

So, 2 pixels belong to the vehicle region.

5. Applying Level Set Update

A level set function evolves over iterations to fit the boundary of an object. At one pixel, the current level set value is $\phi = 6$. The update term is -2 , meaning the contour moves closer to that pixel. Apply the update formula and find the new level set value.

- A. 2
- B. 3
- C. 4
- D. 8

Correct Answer: C. 4

Explanation:

The new level set value is calculated as:

$$\phi_{new} = \phi_{old} + Update$$

$$\begin{aligned}\phi_{new} &= 6 + (-2) \\ \phi_{new} &= 4\end{aligned}$$

So, the new level set value is 4.

6. Applying Area Estimation from Segmented Region

A level set segmentation system detects a diseased region in a plant leaf image. After convergence, the inside region is represented by negative level set values. The system counts 950 pixels with $\phi < 0$. Each pixel represents 0.2 mm^2 in the actual leaf image. The system calculates the diseased area. Apply area calculation and find the diseased region area.

- A. 150 mm^2
- B. 180 mm^2
- C. 190 mm^2
- D. 220 mm^2

Correct Answer: C. 190 mm^2

Explanation:

$$\begin{aligned}Area &= \text{Number of Pixels} \times \text{Area per Pixel} \\ Area &= 950 \times 0.2 = 190 \text{ mm}^2\end{aligned}$$

So, the diseased region area is 190 mm^2 .

1. Applying Fourier Descriptor for Boundary Point Representation

A traffic sign recognition system extracts the contour of a circular sign. One boundary point has coordinate $(x, y) = (6, 8)$. The system converts this point into complex form before applying Fourier descriptors. Apply the complex boundary representation and identify the correct value of $z(k)$.

- A. $6 + 8j$
- B. $8 + 6j$
- C. $14 + j$
- D. $48j$

Correct Answer: A. $6 + 8j$

Explanation:

The boundary points are represented as complex numbers using:

$$z(k) = x(k) + jy(k)$$

$$z(k) = x(k) + jy(k)$$

Given:

$$\begin{aligned}x &= 6, y = 8 \\z(k) &= 6 + 8j\end{aligned}$$

So, the boundary point is represented as $6 + 8j$.

2. Applying Magnitude of a Fourier Descriptor

An object recognition system uses Fourier descriptors to represent the shape of an object boundary. One Fourier descriptor coefficient is given as: $F = 3 + 4j$. The system calculates the magnitude of this coefficient to use it as a shape feature. Apply the magnitude formula and find the descriptor magnitude.

- A. 3
- B. 4
- C. 5
- D. 7

Correct Answer: C. 5

Explanation:

$$\begin{aligned}|F| &= \sqrt{a^2 + b^2} \\|F| &= \sqrt{3^2 + 4^2} \\|F| &= \sqrt{9 + 16} = \sqrt{25} = 5\end{aligned}$$

So, the descriptor magnitude is 5.

3. Applying Low-Frequency Fourier Descriptor Selection

A shape matching system extracts 40 Fourier descriptor coefficients from an object contour. To reduce noise and retain only the main shape information, it keeps only 25% of the lowest-frequency coefficients. These low-frequency descriptors preserve the general object shape. Apply the percentage calculation and find the number of coefficients retained.

- A. 5
- B. 10
- C. 15
- D. 20

Correct Answer: B. 10

Explanation:

$$\begin{aligned} \text{Retained Coefficients} &= 40 \times \frac{25}{100} \\ \text{Retained Coefficients} &= 40 \times 0.25 = 10 \end{aligned}$$

So, 10 coefficients are retained.

4. Applying Wavelet Approximation Coefficient Count

A medical image analysis system applies a one-level 2D wavelet transform to a grayscale image of size 256×256 . After one-level decomposition, the approximation sub-band has half the width and half the height of the original image. The system uses this approximation band for coarse image representation. Apply the wavelet decomposition rule and find the size of the approximation sub-band.

- A. 256×256
- B. 256×128
- C. 128×256
- D. 128×128

Correct Answer: D. 128×128

Explanation:

In one-level 2D wavelet decomposition, the image is downsampled by 2 in both directions.

$$\begin{aligned} \text{Width} &= \frac{256}{2} = 128 \\ \text{Height} &= \frac{256}{2} = 128 \end{aligned}$$

So, the approximation sub-band size is 128×128 .

5. Applying Total Number of Wavelet Coefficients in Sub-Bands

A satellite image of size 128×128 is decomposed using one-level 2D wavelet transform. The result contains four sub-bands: LL, LH, HL, and HH. Each sub-band has size 64×64 . The system wants to calculate the total number of coefficients across all four sub-bands. Apply wavelet coefficient counting and find the total number of coefficients.

- A. 4,096
- B. 8,192
- C. 12,288
- D. 16,384

Correct Answer: D. 16,384

Explanation:

Each sub-band contains:

$$64 \times 64 = 4096$$

There are four sub-bands:

$$Total = 4096 \times 4 = 16384$$

So, the total number of coefficients is 16,384.

6. Applying Energy Calculation of Wavelet Coefficients

A texture classification system extracts wavelet coefficients from a fabric image. Four coefficients from a detail sub-band are: 2, -3 , 4, -1 . The wavelet energy is calculated as the sum of squared coefficient values. Apply the energy formula and find the wavelet energy.

- A. 20
- B. 25
- C. 30
- D. 35

Correct Answer: C. 30

Explanation:

$$\begin{aligned} Energy &= 2^2 + (-3)^2 + 4^2 + (-1)^2 \\ Energy &= 4 + 9 + 16 + 1 = 30 \end{aligned}$$

So, the wavelet energy is 30.

2. Applying Skeleton Length Calculation

A medical image analysis system uses medial representation to study a blood vessel. The extracted vessel skeleton contains 5 connected line segments. Each segment has a length of 12 pixels. The system calculates the total skeleton length to estimate vessel structure. Apply skeleton length calculation and find the total length.

- A. 48 pixels
- B. 60 pixels
- C. 72 pixels
- D. 84 pixels

Correct Answer: B. 60 pixels

Explanation:

Total Skeleton Length = Number of Segments $\times$ Length per Segment

Total Skeleton Length = $5 \times 12 = 60$

So, the total skeleton length is 60 pixels.

3. Applying Radius in Medial Axis Representation

A shape analysis system represents an object using medial points. One medial point is located at the center of a circular object. The distance from this medial point to the object boundary is 25 pixels. This distance is treated as the radius of the maximal inscribed circle at that medial point. Apply medial representation and identify the radius value.

- A. 10 pixels
- B. 15 pixels
- C. 20 pixels
- D. 25 pixels

Correct Answer: D. 25 pixels

Explanation:

In medial representation, each medial point can be associated with a radius. This radius is the distance from the medial point to the nearest boundary.

Radius=Distance from medial point to boundary

So, the radius is 25 pixels.

4. Applying Diameter from Medial Radius

A computer vision system analyzes a circular cell using medial representation. The medial point is at the center of the cell, and the radius from the medial point to the boundary is 18 pixels. The system wants to estimate the diameter of the cell. Apply the diameter formula and calculate the cell diameter.

A. 18 pixels

B. 24 pixels

C. 36 pixels

D. 48 pixels

Correct Answer: C. 36 pixels

Explanation:

Diameter=2×Radius

Diameter=2×18=36

So, the estimated diameter is 36 pixels.

5. Applying Branch Count in Skeleton Representation

A leaf shape is represented using a skeleton structure. The main skeleton branch has 3 side branches on the left and 4 side branches on the right. These branches represent the structural pattern of the leaf. The system counts the total number of side branches for classification. Apply branch counting and find the total number of side branches.

- A. 5
- B. 6
- C. 7
- D. 8

Correct Answer: C. 7

Explanation:

Total Side Branches=Left Branches+Right Branches

Total Side Branches=3+4=7

So, the skeleton has 7 side branches.

6. Applying Average Radius in Medial Representation

A vessel analysis system extracts medial points from a blood vessel region. The radii measured at four medial points are 4, 6, 8, and 10 pixels. The system calculates the average radius to describe the vessel thickness. Apply the average formula and find the average medial radius.

- A. 6 pixels
- B. 7 pixels
- C. 8 pixels
- D. 9 pixels

Correct Answer: B. 7 pixels

Explanation:

Formula:

$$\begin{aligned} \text{Average Radius} &= \frac{\text{Sum of Radii}}{\text{Number of Points}} \\ &= \frac{4 + 6 + 8 + 10}{4} \\ &= \frac{28}{4} = 7 \end{aligned}$$

So, the average medial radius is **7 pixels**.

1. Applying Image Downsampling in Multi-Resolution Analysis

A satellite image processing system uses multi-resolution analysis to study agricultural land. The original image size is 512×512 pixels. At the next lower resolution level, the image is downsampled by a factor of 2 in both width and height. This lower-resolution image is used to observe large crop regions with less detail. Apply the downsampling rule and find the size of the lower-resolution image.

- A. 512×256
- B. 256×512
- C. 256×256
- D. 128×128

Correct Answer: C. 256×256

Explanation:

$$\begin{aligned} \text{New Width} &= \frac{512}{2} = 256 \\ \text{New Height} &= \frac{512}{2} = 256 \end{aligned}$$

So, the lower-resolution image size is:

$$256 \times 256$$

2. Applying Pixel Count at Lower Resolution

A medical image analysis system creates a multi-resolution pyramid of an MRI image. The original image has size 1024×1024 pixels. After one level of downsampling by a factor of 2 in both directions, the system calculates the number of pixels in the reduced image. This helps reduce computational cost during coarse analysis. Apply the pixel count calculation and find the number of pixels after downsampling.

- A. 131,072 pixels
- B. 262,144 pixels

- C. 524,288 pixels
- D. 1,048,576 pixels

Correct Answer: B. 262,144 pixels

Explanation:

After downsampling:

$$\begin{aligned} \text{New Size} &= 512 \times 512 \\ \text{Total Pixels} &= 512 \times 512 = 262144 \end{aligned}$$

So, the reduced image contains 262,144 pixels.

3. Applying Multi-Level Image Pyramid Size

A computer vision system uses an image pyramid for object detection. The original image size is 800×600 pixels. The system reduces the image by half at each level. After one level, the size becomes 400×300 . After the second level, it is again reduced by half. Apply the multi-resolution pyramid rule and find the image size after two levels.

- A. 400×300
- B. 300×200
- C. 200×150
- D. 100×75

Correct Answer: C. 200×150

Explanation:

Original image:

$$800 \times 600$$

After level 1:

$$400 \times 300$$

After level 2:

$$200 \times 150$$

So, the image size after two levels is 200×150 .

4. Applying Wavelet Sub-Band Size in Multi-Resolution Analysis

A texture classification system applies one-level 2D wavelet decomposition to a fabric image of size 256×256 pixels. The decomposition produces four sub-bands: LL, LH, HL, and HH. Each sub-band has

half the width and half the height of the original image. The LL band is used for coarse approximation. Apply the wavelet decomposition rule and find the size of each sub-band.

- A. 256×256
- B. 256×128
- C. 128×128
- D. 64×64

Correct Answer: C. 128×128

Explanation:

$$\begin{aligned} \text{Subband Width} &= \frac{256}{2} = 128 \\ \text{Subband Height} &= \frac{256}{2} = 128 \end{aligned}$$

So, each sub-band size is:

$$128 \times 128$$

5. Applying Total Coefficients in Wavelet Multi-Resolution

A remote sensing image of size 128×128 pixels is decomposed using one-level wavelet transform. It produces four sub-bands: LL, LH, HL, and HH. Each sub-band has size 64×64 . The system calculates the total number of coefficients across all sub-bands. Apply coefficient counting and find the total number of coefficients.

- A. 4,096
- B. 8,192
- C. 12,288
- D. 16,384

Correct Answer: D. 16,384

Explanation:

Each sub-band contains:

$$64 \times 64 = 4096$$

There are four sub-bands:

$$\text{Total} = 4096 \times 4 = 16384$$

So, the total number of coefficients is 16,384.

6. Applying Scale Reduction Factor

A surveillance system uses multi-resolution analysis to detect objects at different sizes. The original image width is 640 pixels. At each pyramid level, the width is reduced by half. The engineer wants to know the width at the third level after three reductions. Apply scale reduction and find the width after three levels.

- A. 80 pixels
- B. 160 pixels
- C. 240 pixels
- D. 320 pixels

Correct Answer: A. 80 pixels

Explanation:

Original width:

640

After level 1:

320

After level 2:

160

After level 3:

80

So, the width after three reductions is 80 pixels.